

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama, Belgaum – 590014



Report on

“Disaster Response System using Social Media Data”

Project Based Assignment

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Submitted by

Pranav Kumar S A	1AY23CD045
Pratap H	1AY23CD046
Prateek Kumar Bal	1AY23CD047
Pratheek R S	1AY23CD048

Under the guidance of

Ms. Surbhi

Assistant Professor

Department of AIML and CSE (DS)

Department of

**Computer Science Engineering
(Data Science)**

Acharya Institute of Technology

(An Institute Affiliated to Visvesvaraya Technological University, Belgaum, Recognized by AICTE)

Acharya Dr. Sarvepalli Radhakrishnan Road, Soladevanahalli, Bangalore-560107

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Abstract

The rapid growth of social media platforms has enabled people to share information about disasters and emergencies in real time. During natural disasters such as floods, earthquakes, cyclones, and fires, social media data can provide valuable insights for emergency response and disaster management. However, handling and analyzing such large-scale streaming data using traditional systems is difficult due to high data volume, velocity, and variability.

This project presents a Big Data based Disaster Response System using social media data. The system is designed to detect disaster-related information from social media streams and generate alerts for affected regions. A disaster tweet dataset was used to simulate real-time social media streaming. Hadoop Distributed File System (HDFS) was used for distributed storage of tweet logs, while Apache Spark was utilized for real-time processing and filtering of disaster-related content. Hive was integrated for analytical querying within the Hadoop ecosystem, and MongoDB was used to store generated disaster alerts in a flexible NoSQL format. Link analytics and graph visualization techniques were also implemented to analyze relationships between disasters and affected locations.

Introduction

Natural disasters such as floods, earthquakes, cyclones, storms, and fires cause significant damage to human life, infrastructure, and the environment every year. Rapid detection and timely response during such events are essential to minimize losses and improve disaster management efficiency. In recent years, social media platforms have emerged as important sources of real-time information during emergencies, as users frequently share updates, images, and reports about ongoing disaster situations.

The large volume and continuous flow of social media data make traditional data processing systems inefficient for real-time analysis. Big Data technologies provide scalable and distributed solutions capable of handling massive datasets with high speed and reliability. Technologies such as Hadoop, Spark, Hive, and MongoDB enable efficient storage, processing, querying, and analysis of large-scale streaming data.

This project focuses on developing a Disaster Response System using social media data and Big Data technologies. Apache Spark is used for real-time stream processing, while MongoDB stores generated alerts for quick access and analysis.

System Architecture

- The system is designed using a distributed Big Data architecture for handling large-scale disaster-related social media data.
- Disaster tweet data is used to simulate real-time social media streaming. Apache Spark Streaming is used for continuous real-time processing of incoming tweet data.
- Disaster-related tweets are filtered using keyword-based distributed processing techniques. Hadoop Distributed File System (HDFS) is used for scalable and fault-tolerant storage of tweet logs.
- Distributed processing concepts based on the MapReduce paradigm are applied for filtering relevant disaster information.
- Hive is integrated to support SQL-like analytical querying on disaster data stored within the Hadoop ecosystem. MongoDB is used as a NoSQL database to store generated disaster alerts in JSON document format. Link analytics and graph visualization techniques are used to analyze relationships between disasters and affected locations.

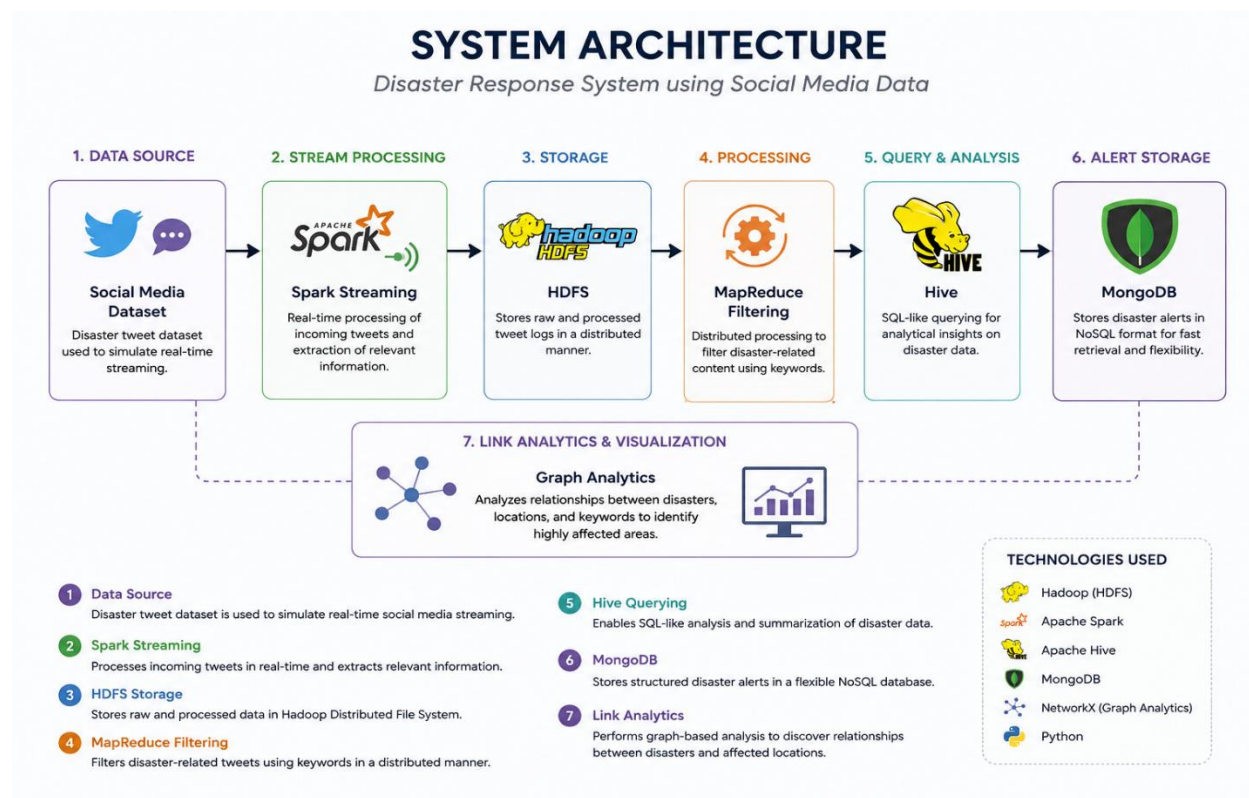


Fig 1. System Architecture

Technologies Used

Table 1. Software and tools Used

Technology / Tool	Purpose in the Project
Apache Hadoop	Provides the Big Data framework for distributed storage and processing.
HDFS (Hadoop Distributed File System)	Stores disaster tweet logs in a scalable and fault-tolerant distributed environment.
Apache Spark	Performs real-time stream processing and disaster tweet analysis.
Spark Streaming	Simulates continuous real-time tweet processing and alert generation.
Apache Hive	Supports SQL-like querying and analytical processing of disaster-related data.
MongoDB	Stores generated disaster alerts in flexible NoSQL document format.
Python	Used for scripting, data processing, Spark integration, and analytics implementation.
PySpark	Enables Python integration with Apache Spark for distributed processing.
NetworkX	Performs graph-based link analytics between disasters and affected locations.
Matplotlib	Generates graphical visualizations and analytics charts.
WSL (Windows Subsystem for Linux)	Provides Linux environment for running Hadoop ecosystem tools on Windows. Development and execution environment for scripts and commands.
Kaggle Disaster Tweets Dataset	Dataset used to simulate real-time social media disaster data.

Dataset Description

The dataset used in this project is the “Natural Language Processing with Disaster Tweets” dataset obtained from Kaggle. The dataset contains disaster-related and non-disaster-related tweets collected from social media platforms. It is commonly used for disaster detection and text classification tasks.

The dataset is used to simulate real-time disaster tweet streaming for processing and analysis using Big Data technologies.

Table 2. Dataset Features

Attribute	Description
id	Unique identifier for each tweet
keyword	Disaster-related keyword associated with the tweet
location	Location information mentioned in the tweet
text	Actual tweet content
target	Indicates whether the tweet is related to a real disaster (1) or not (0)

Dataset Characteristics

- The dataset contains thousands of disaster-related tweets.
- Tweets include information about floods, earthquakes, fires, cyclones, storms, and other emergency events.
- Both relevant and irrelevant tweets are included, making the dataset suitable for filtering and classification tasks.
- The dataset is semi-structured and text-intensive in nature.

Source

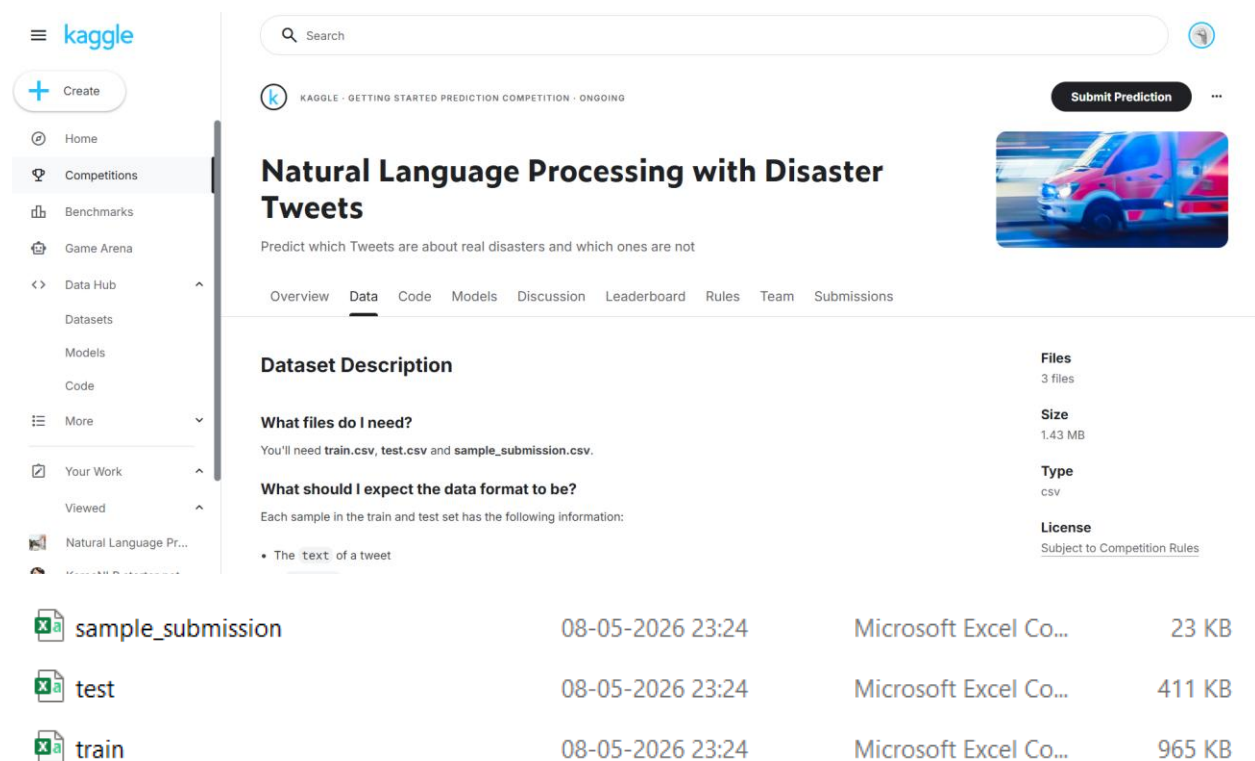
Kaggle – Natural Language Processing with Disaster Tweets Dataset.

Methodology (Implementation)

The proposed Disaster Response System follows a step-by-step Big Data processing workflow for handling disaster-related social media data. Multiple Big Data technologies are integrated to perform distributed storage, real-time processing, analytical querying, alert generation, and visualization.

1. Data Collection

A disaster tweet dataset obtained from Kaggle is used as the primary data source. The dataset contains tweets related to disasters such as floods, earthquakes, fires, cyclones, and storms. The dataset is used to simulate real-time social media streaming.



The screenshot displays the Kaggle dataset page for "Natural Language Processing with Disaster Tweets". The page includes a search bar, a "Submit Prediction" button, and a "Dataset Description" section. The description states: "Predict which Tweets are about real disasters and which ones are not". It also lists the files: "train.csv", "test.csv", and "sample_submission.csv". The "What files do I need?" section mentions: "You'll need train.csv, test.csv and sample_submission.csv." The "What should I expect the data format to be?" section states: "Each sample in the train and test set has the following information: The text of a tweet". The "Files" section lists: "3 files". The "Size" section lists: "1.43 MB". The "Type" section lists: "csv". The "License" section lists: "Subject to Competition Rules". Below the description, there is a table with the following data:

File Name	Created At	File Type	Size
sample_submission	08-05-2026 23:24	Microsoft Excel Co...	23 KB
test	08-05-2026 23:24	Microsoft Excel Co...	411 KB
train	08-05-2026 23:24	Microsoft Excel Co...	965 KB

Fig 2. Dataset Description

2. Real-Time Stream Processing using Spark

Apache Spark Streaming is used to process incoming tweet data continuously. Tweets are read one-by-one to simulate live social media feeds. Disaster-related keywords are identified using keyword filtering techniques.

Keywords used for filtering include:

- Flood
- Earthquake
- Fire
- Cyclone
- Storm

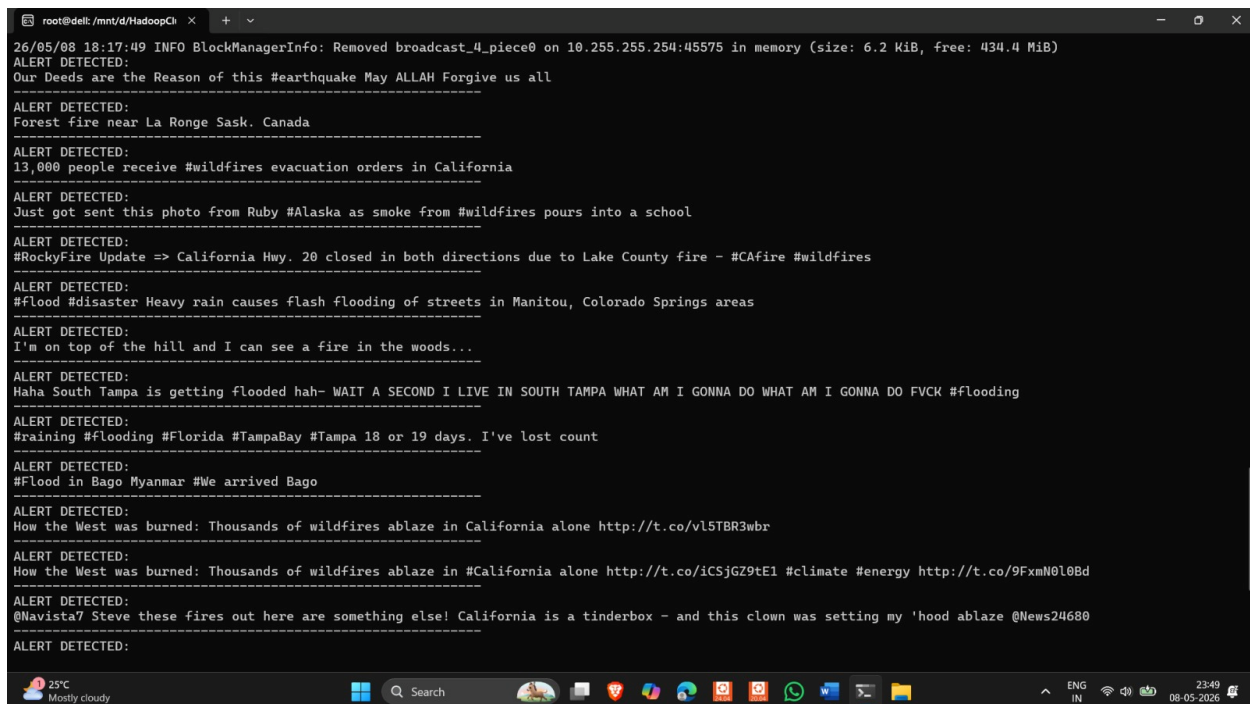
A screenshot of a PySpark terminal window. The window title is 'root@deli: /mnt/d/HadoopCli'. The terminal displays a series of log messages from a stream processing application. The logs include timestamps, memory usage information, and several 'ALERT DETECTED:' messages. These alerts contain text extracted from tweets, such as 'Our Deeds are the Reason of this #earthquake May ALLAH Forgive us all', 'Forest fire near La Ronge Sask. Canada', '13,000 people receive #wildfires evacuation orders in California', and 'Just got sent this photo from Ruby #Alaska as smoke from #wildfires pours into a school'. The terminal also shows some user input and output, like 'Haha South Tampa is getting flooded hah- WAIT A SECOND I LIVE IN SOUTH TAMPA WHAT AM I GONNA DO WHAT AM I GONNA DO FVCK #flooding'. The bottom of the window shows a Windows taskbar with various icons and system information like '25°C Mostly cloudy' and '23:49 08-05-2026'.

Fig 3. Stream Processing using PySpark

3. HDFS Storage

The disaster tweet dataset and processed logs are stored in Hadoop Distributed File System (HDFS). HDFS provides distributed and fault-tolerant storage for large-scale data processing.

The dataset is uploaded into HDFS directories for distributed access and management.

4. Distributed Filtering using MapReduce Concepts

Relevant disaster tweets are filtered using distributed processing logic following the MapReduce paradigm. Tweets containing disaster-related keywords are identified and processed for alert generation and analysis.


```
root@dell:~# cd /mnt/HadoopClusternlp-getting-started
-bash: cd: /mnt/HadoopClusternlp-getting-started: No such file or directory
root@dell:~# cd /mnt/HadoopCluster/nlp-getting-started
-bash: cd: /mnt/HadoopCluster/nlp-getting-started: No such file or directory
root@dell:~# cd /mnt/d/HadoopCluster/nlp-getting-started
root@dell:/mnt/d/HadoopCluster/nlp-getting-started# hdfs dfs -put train.csv /disaster_data
2026-05-08 17:56:19,990 WARN util.NativeCodeLoader: Unable to load native-hadoop library for
root@dell:/mnt/d/HadoopCluster/nlp-getting-started# hdfs dfs -ls /disaster_data
2026-05-08 17:57:19,877 WARN util.NativeCodeLoader: Unable to load native-hadoop library for
Found 1 items
-rw-r--r-- 1 root supergroup 987712 2026-05-08 17:56 /disaster_data/train.csv
root@dell:/mnt/d/HadoopCluster/nlp-getting-started# |
```

Fig 4. HDFS directory disaster_data containing train.csv

5. Hive Integration and Querying

Apache Hive is integrated into the Hadoop ecosystem to support SQL-like querying and analytical operations on disaster-related data.

Hive enables structured analysis and summarization of tweet information stored in the distributed environment.

```
SLF4J: Found binding in [jar:file:/root/hadoop/share/hadoop/common/lib/slf4j-reload4j-1.7.36.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
Hive Session ID = 6elac7df-bffb-466a-8ab0-289e54eae4fe

Logging initialized using configuration in jar:file:/root/hive/lib/hive-common-3.1.3.jar!/hive-log4j2.properties Async: true
Hive Session ID = 438808dc-19c8-4f72-8ca0-8d2f7ff212a0
Hive-on-MR is deprecated in Hive 2 and may not be available in the future versions. Consider using a different execution engine (i.e. spark, tez) or using H
ive 1.X releases.
hive> CREATE DATABASE disasterDB;
OK
Time taken: 0.541 seconds
hive> USE disasterDB;
OK
Time taken: 0.046 seconds
hive> CREATE TABLE disaster_tweets (
  > id STRING,
  > keyword STRING,
  > location STRING,
  > text STRING,
  > target INT,
  > )
  > ROW FORMAT DELIMITED
  > FIELDS TERMINATED BY ','
  > STORED AS TEXTFILE
  > TBLPROPERTIES ("skip.header.line.count"="1");
OK
Time taken: 0.745 seconds
hive> LOAD DATA LOCAL INPATH '/mnt/d/HadoopCluster/nlp-getting-started/train.csv'
  > INTO TABLE disaster_tweets;
Loading data to table disasterdb.disaster_tweets
OK
Time taken: 1.359 seconds
hive> SELECT * FROM disaster_tweets LIMIT 5;
OK
1      Our Deeds are the Reason of this #earthquake May ALLAH Forgive us all 1
4      Forest fire near La Ronge Sask. Canada 1
5      ALL residents asked to 'shelter in place' are being notified by officers. No other evacuation or shelter in place orders are expecte
d      1
6      "13 NULL
7      Just got sent this photo from Ruby #Alaska as smoke from #wildfires pours into a school 1
```

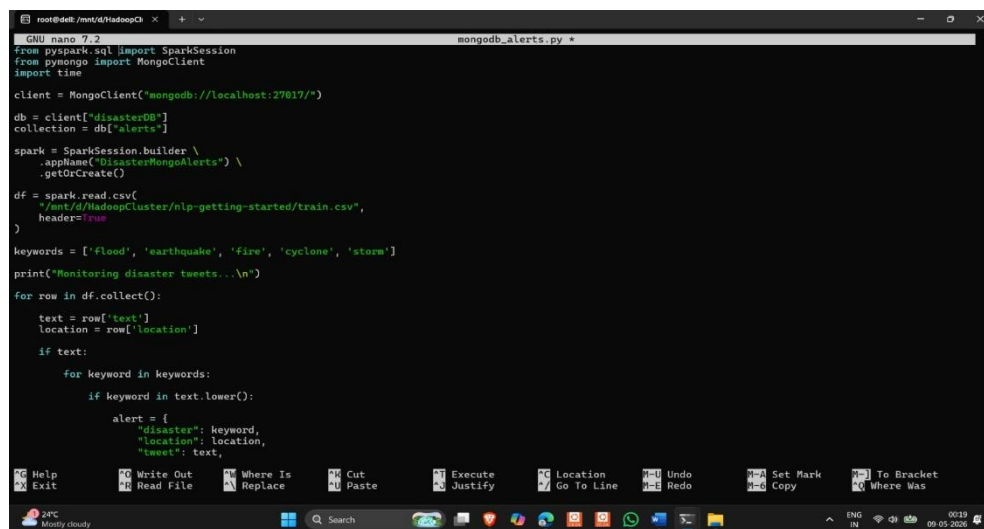
Fig 5. Hive Queries for tweet analysis and summarization

6. MongoDB Alert Storage

Filtered disaster alerts are stored in MongoDB using JSON document format. Each alert contains:

- Disaster type
- Location
- Severity level
- Tweet content

MongoDB provides flexible NoSQL storage for semi-structured real-time alert data.



```
GNU nano 7.2 mongodb_alerts.py
from pyspark.sql import SparkSession
from pymongo import MongoClient
import time

client = MongoClient("mongodb://localhost:27017/")
db = client["disasterDB"]
collection = db["alerts"]

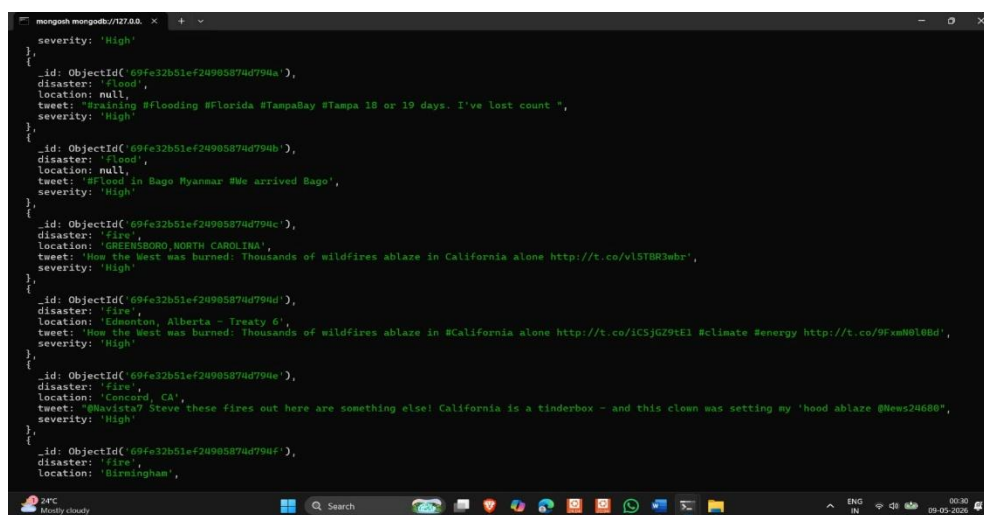
spark = SparkSession.builder \
    .appName("DisasterMongoAlerts") \
    .getOrCreate()

df = spark.read.csv(
    "/mnt/d/HadoopCluster/nlp-getting-started/train.csv",
    header=True
)

keywords = ['flood', 'earthquake', 'fire', 'cyclone', 'storm']

print("Monitoring disaster tweets...\n")
for row in df.collect():
    text = row['text']
    location = row['location']
    if text:
        for keyword in keywords:
            if keyword in text.lower():
                alert = {
                    "disaster": keyword,
                    "location": location,
                    "tweet": text,
                    "severity": "High"
                }
```

Fig 6. Mongo_alerts.py for save alerts in MongoDB Collections



```
severity: 'High'
},
{
  _id: ObjectId('69fe32b51ef24985874d794a'),
  disaster: 'flood',
  location: null,
  tweet: 'Raining #flooding #Florida #TampaBay #Tampa 18 or 19 days. I've lost count ',
  severity: 'High'
},
{
  _id: ObjectId('69fe32b51ef24985874d794b'),
  disaster: 'flood',
  location: null,
  tweet: '#Flood in Bago Myanmar #We arrived Bago',
  severity: 'High'
},
{
  _id: ObjectId('69fe32b51ef24985874d794c'),
  disaster: 'fire',
  location: 'GREENSBORO, NORTH CAROLINA',
  tweet: 'How the West was burned: Thousands of wildfires ablaze in California alone http://t.co/v15TRK3ubr',
  severity: 'High'
},
{
  _id: ObjectId('69fe32b51ef24985874d794d'),
  disaster: 'fire',
  location: 'Edmonton, Alberta - Treaty 6',
  tweet: 'Hop the West was burned: Thousands of wildfires ablaze in #California alone http://t.co/1CSJGZ9tE1 #Climate #energy http://t.co/9FXmN010Bd',
  severity: 'High'
},
{
  _id: ObjectId('69fe32b51ef24985874d794e'),
  disaster: 'fire',
  location: 'Concord, CA',
  tweet: '@Bavixty? Steve these fires out here are something else! California is a tinderbox - and this clown was setting my hood ablaze @News24688',
  severity: 'High'
},
{
  _id: ObjectId('69fe32b51ef24985874d794f'),
  disaster: 'fire',
  location: 'Birmingham',
  severity: 'High'
}
```

Fig 7. Alerts saved in disasterDB Collection

7. Link Analytics and Visualization

Graph-based link analytics is performed using NetworkX and Matplotlib libraries. Relationships between disasters and affected locations are visualized using graph structures to identify highly impacted regions and disaster patterns.

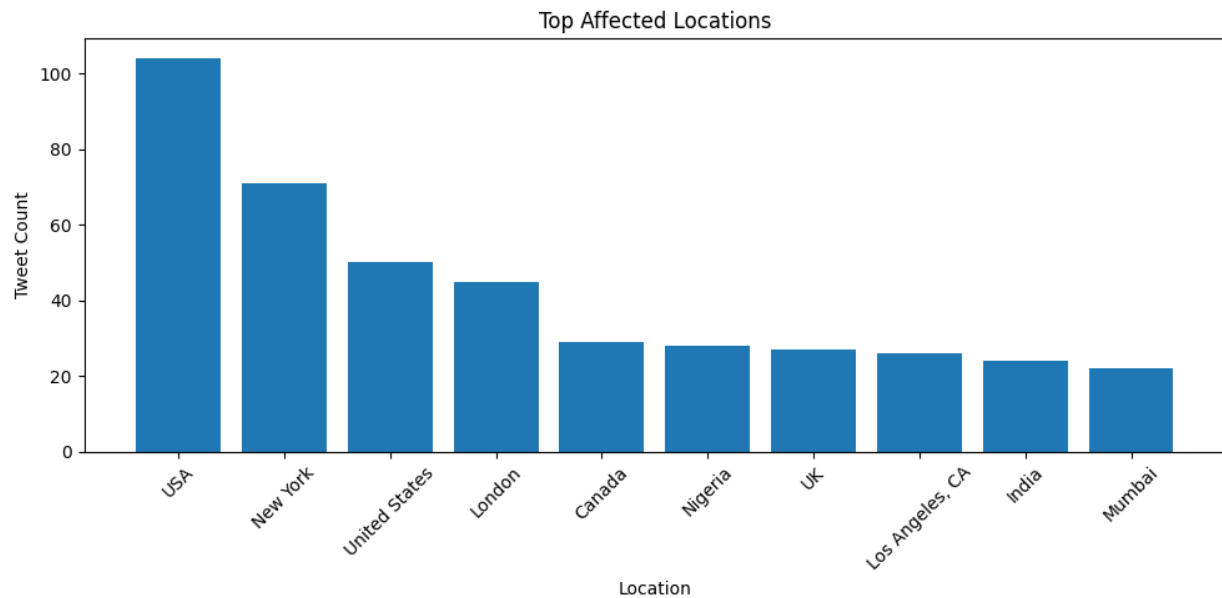


Fig 8. Tweets from various countries

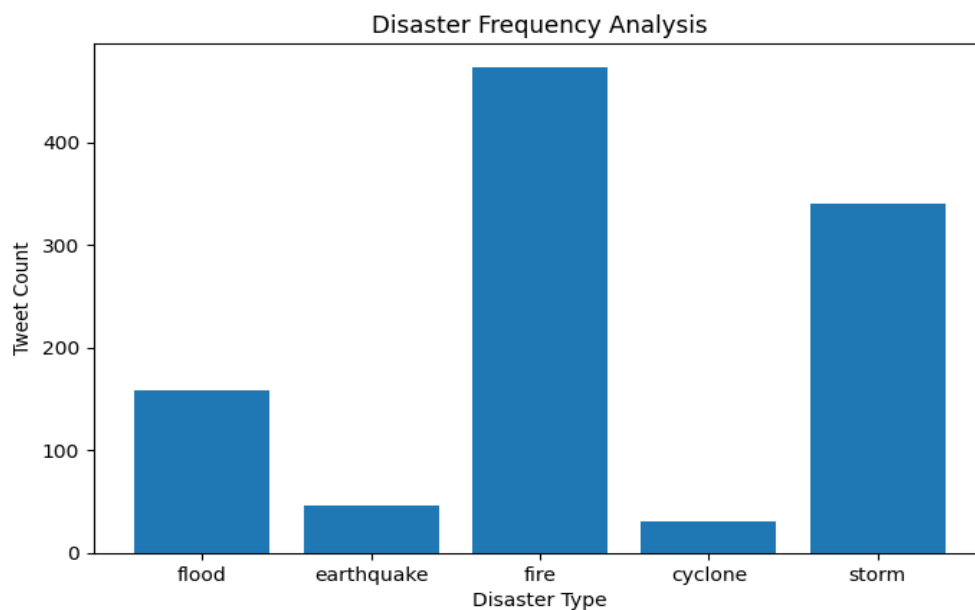


Fig 9. Frequency (occurrences) of different disaster types

Results

The Disaster Response System using Social Media Data was successfully implemented using multiple Big Data technologies including Hadoop, Spark, Hive, MongoDB, and graph analytics tools. The system was capable of processing disaster-related tweet data, generating alerts, and visualizing affected locations.

- Disaster tweets were successfully processed using Apache Spark Streaming.
- Disaster-related tweets were filtered using keyword-based analysis.
- Tweet logs were stored in Hadoop Distributed File System (HDFS).
- Disaster alerts were stored in MongoDB using JSON format.
- Link analytics and multiple visualizations were generated successfully.

Table 3. Advantages, Limitations and Future Enhancements

Advantages	Limitations	Future Enhancements
Supports scalable distributed storage and processing of large-scale social media data.	Uses a pre-collected dataset instead of live social media APIs.	Integrate live Twitter/X API for real-time disaster data collection.
Enables near real-time disaster monitoring and alert generation.	Real-time streaming is simulated rather than fully live.	Implement advanced NLP and machine learning models for intelligent disaster detection.
Provides efficient stream processing using Apache Spark Streaming.	Implemented on a single-node Hadoop environment.	Add sentiment analysis and severity prediction mechanisms.
Supports flexible NoSQL storage using MongoDB.	Disaster detection is primarily keyword-based.	Develop interactive dashboards and GIS-based heatmap visualizations.
Integrates multiple Big Data technologies into a unified analytics pipeline.	Hive metastore instability may occur in WSL environments.	Deploy the system on cloud platforms for scalable real-world applications.

Conclusion

The “Disaster Response System using Social Media Data” successfully demonstrated the practical implementation of a Big Data based disaster monitoring and analytics pipeline using modern distributed technologies. The project focused on processing disaster-related social media information efficiently through scalable storage, real-time stream processing, analytical querying, NoSQL alert management, and graph-based visualization techniques.

With the rapid growth of social media platforms, disaster-related information is generated continuously in massive volumes. Traditional systems often face challenges in processing such large-scale streaming data efficiently and in real time. The proposed system addressed these challenges by integrating multiple Big Data technologies including Hadoop, HDFS, Spark Streaming, Hive, MongoDB, and graph analytics tools into a unified architecture capable of handling disaster-related tweet data effectively.

Apache Hadoop and HDFS were used to provide scalable and fault-tolerant distributed storage for disaster tweet logs. Apache Spark Streaming enabled near real-time tweet processing and disaster keyword filtering, allowing the system to simulate continuous disaster monitoring. Hive and Spark SQL were integrated to support analytical querying and structured data analysis within the Hadoop ecosystem. MongoDB provided flexible NoSQL storage for disaster alerts generated during processing, while graph analytics and visualization techniques were used to identify relationships between disasters and affected locations.

The project also demonstrated the importance of data analytics and visualization in disaster response systems. Multiple analytical outputs including bar charts, pie charts, trend graphs, word clouds, and link analytics graphs were generated to provide meaningful insights into disaster patterns, frequently affected regions, and disaster-related social media activity. These visualizations improved the interpretability of the processed data and enhanced overall analytical capabilities.

Overall, the proposed system successfully achieved its objectives and demonstrated the effectiveness of integrating distributed computing, stream processing, NoSQL databases, and analytical visualization techniques for real-time disaster monitoring and response analysis using social media data. The project further emphasizes the growing role of Big Data technologies in solving real-world emergency management and disaster response challenges.

Team Member Contributions

Table 4. Individual Contributions

Team Member	Contribution
Pranav	Configured Hadoop ecosystem, managed HDFS setup, uploaded dataset to distributed storage, and handled Hadoop cluster operations.
Pratap	Implemented Apache Spark Streaming for real-time disaster tweet processing and keyword-based filtering mechanisms.
Prateek	Integrated MongoDB for disaster alert storage and managed JSON-based NoSQL database operations.
Pratheek	Developed link analytics visualizations, generated analytical graphs and charts, and prepared project documentation and presentation materials.

Collaborative Contributions

- All team members participated in system design, testing, debugging, and integration of Big Data technologies.
- Team members collectively contributed to report preparation, result analysis, screenshot collection, and final project presentation.
- The project was developed through coordinated teamwork involving distributed storage, real-time processing, database management, and analytical visualization modules.

Team Coordination and Workflow

- Work was divided based on individual technical modules to ensure efficient project development.
- Regular discussions and integration sessions were conducted to combine all components into a unified disaster response pipeline.
- All modules were tested collectively to ensure smooth communication between Hadoop, Spark, MongoDB, and analytics components.

References

1. Apache Hadoop Documentation. Apache Software Foundation. Available at: <https://hadoop.apache.org/docs/>
Accessed for understanding Hadoop ecosystem configuration, HDFS architecture, distributed storage mechanisms, and cluster management concepts used in the project.
2. Apache Spark Documentation. Apache Software Foundation. Available at: <https://spark.apache.org/documentation.html>
Referred for implementing Spark Streaming, real-time data processing workflows, and distributed analytical operations using PySpark.
3. Apache Hive Documentation. Apache Software Foundation. Available at: <https://hive.apache.org/>
Used for studying Hive architecture, SQL-like querying mechanisms, schema management, and analytical querying integration within the Hadoop ecosystem.
4. MongoDB Official Documentation. MongoDB Inc. Available at: <https://www.mongodb.com/docs/>
Referred for implementing NoSQL-based disaster alert storage using JSON document structures and database query operations.
5. PySpark API Documentation. Apache Spark. Available at: <https://spark.apache.org/docs/latest/api/python/>
Used for understanding Python integration with Apache Spark and implementing distributed stream processing operations.
6. Kaggle – Natural Language Processing with Disaster Tweets Dataset. Available at: <https://www.kaggle.com/competitions/nlp-getting-started>
The primary dataset source used for simulating real-time disaster-related social media tweet streams and performing analytical processing.
7. NetworkX Documentation. Available at: <https://networkx.org/documentation/stable/>
Used for implementing graph-based link analytics and visualization of relationships between disaster events and affected regions.